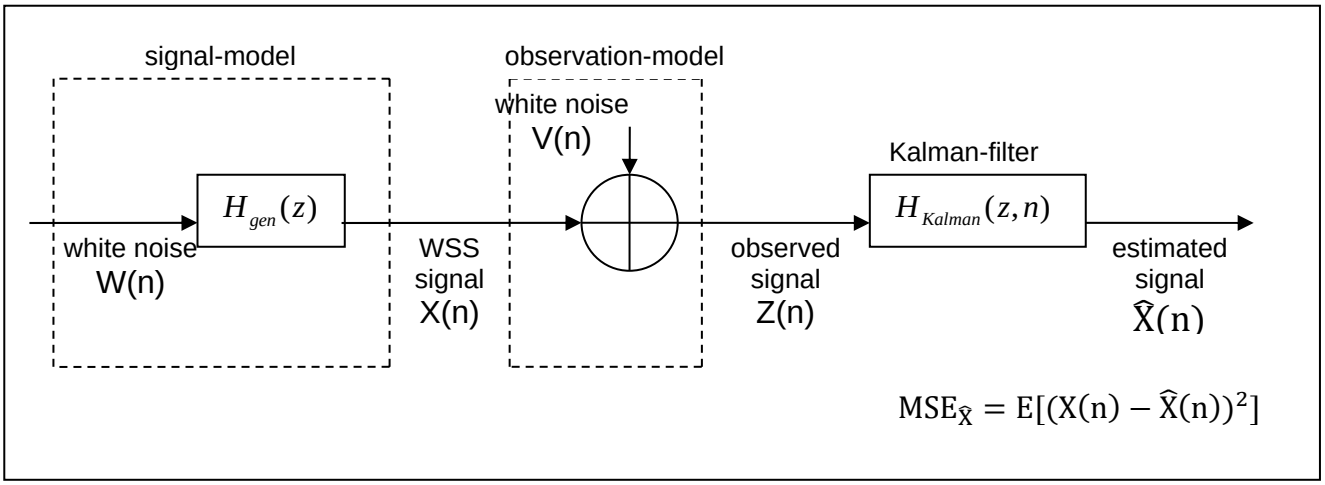
Problem 5.1 (Theoretical + MATLAB)
Signal estimation with discrete-time scalar current Kalman filter

Motivation

This exercise goes through all the most important basic conditions regarding the most applied discrete-time Kalman-filter algorithm – both theoretically and as simulation in MATLAB.
The Kalman filter performs – like the Wiener-filter – stochastic optimal signal estimation (LMMSEE). The difference is that the Kalman-filter is more appropriate for non-stationary situations (e.g. if the noise power varies over time) due to the recursive structure of the algorithm.
Therefore, this exercise will examine both stationary and non-stationary signal models.
Regarding comparisons between theoretical calculations and simulations, the discrete-time case is treated, but all results are absolutely parallel for the continuous-time case.

Signal model for the problem

Below you can see a block diagram for the signal model generating an unobservable random signal, an observation model for measuring the signal and Kalman-filtering of the observed signal with regard to estimation of the unobservable signal.
Note that the first parts of the problem are similar to Problem 4.1 (you may reuse results and MATLAB code), but the **notation** is modified to follow the notation used in the proposed literature for this lecture.



Structure of the exercise

The exercise is divided into the following:

A. Stationary signal and observation-model
B. Steady-state Kalman filter
C. Stationary signal model and non-stationary observation-model

Both theoretical calculations and calculations with MATLAB is to be performed for all three parts.

A. Stationary signal and observation-model

Regarding comparison between Wiener-filter and Kalman-filter, the same signal model as in problem 4.1 is used, i.e.

$$R_{xx}(k) = a^{|k|}, \quad a = 0.8$$
$$W(n) = N\left(0, {\sigma_w}^2(n)\right), \quad {\sigma_w}^2(n) = {\sigma_w}^2$$
$$S_{ww}(\omega) = {\sigma_w}^2$$

1) Write up the signal-model for formation of the signal X(n)
   Remember to specify the model for the "driving noise" signal W(n) also.
   (Use the results from Problem 4.1 question 2+3)

In this part of the problem we also use the same observation model as in Problem 4.1, i.e.

$$V(n) = N\left(0, {\sigma_v}^2(n)\right), \quad {\sigma_v}^2(n) = {\sigma_v}^2 = 4$$
$$S_{vv}(\omega) = {\sigma_v}^2$$

2) Write up the signal-model for formation of the observed signal Z(n)
   Remember to specify the model for the measuring noise V(n) also.
   (Use the results from Problem 4.1 question 2+3)

A current Kalman filter is to be applied for estimation of X(n) from the obs. Z(n)

3) Write up / find the algorithm for the Kalman filter.

4) Formulate the Kalman algorithm as a 1' order IIR filter, i.e. write up
   $$\hat{X}(n) = f(\hat{X}(n-1), Z(n))$$

5) Outline a block diagram for the Kalman IIR-filter, as a canonical type 2 IIR structure is being used.

6) Manually perform the iterations $n = 1$ og $n = 2$ in the algorithm from question 3, as the initial condition X(0) = N(0,1) can be used.

7) MATLAB-realisation:
   - Generate, from the signal model, N=100 samples of W(n) and X(n) (remember ${\sigma_w}^2$)
   - Apply these and the observation model in order to generate N=100 samples of Z(n)
   - Implement the Kalman-filter algorithm and apply it on Z(n)
     (use the same initialization as in question 6)
   - Plot X(n), Z(n), $X_{hat}(n)$, K(n) and P(n)=MSE(n) for the N=100 samples
   - Compare with the manual calculations for $n = 1$ and $n = 2$
   - Comment the generated plots in general.

B. Steady-state Kalman filter

As both the signal model and the observation model in section A of this problem are stationary, the Kalman-filter converges towards a steady-state situation.

8) Write up and solve the steady-state (Ricatti) equation for

$$P_\infty = \lim_{n\to\infty} MSE_{\hat{X}} = \lim_{n\to\infty} E[(X(n) - \hat{X}(n))^2]$$

9) Use this in order to find the steady-state values

$P_\infty^- = \lim_{n\to\infty} MSE_{\hat{X}^-} = \lim_{n\to\infty} E[(X(n) - \hat{X}^-(n))^2]$ and

$K_\infty = \lim_{n\to\infty} K(n)$

Interpret and explain the values of P_∞ and P_∞^-

10) MATLAB-realisation:
- Apply ***dlqe*** for determination of P_∞, P_∞^- and K_∞
- Check with question 8+9

11) Write up the difference equation and the transfer function $H_{Kalman,ss}(z) = \frac{\hat{X}(z)}{Z(z)}$ for the steady-state Kalman-filter.

12) Compare with the non-causal IIR Wiener-filter from Problem 4.1 regarding the transfer function and MSE.
Conclusion? Explanation?

C. Stationary signal model and non-stationary observation-model

Now the observation model in changed to being non-stationary in order to examine the native, "adaptive" characteristics of the Kalman filter.

First, apply the observation model

$$V(n) = N\left(0, {\sigma_v}^2(n)\right), \quad {\sigma_v}^2(n) = 4 + \sin(2\pi\frac{n}{N/2})$$

13) MATLAB-realisation:
- Implement this observation model (other conditions remain unchanged)
- Perform again Kalman-filtering and plots as in question 7.
- Comment and explain.

Then apply the observation model

$$V(n) = N\left(0, {\sigma_v}^2(n)\right), \quad {\sigma_v}^2(n) = 4 \cdot (0.9)^{n-1}$$

14) MATLAB-realisation:
- Implement, plot and explain as in question 13.

15) How should / could the observation models in question 13+14 be handled if you wanted to use a Wiener-filter instead of the Kalman-filter for signal estimation?

Problem 5.2 (Theoretical + MATLAB)
Vektor Kalman filter applied to 1D tracking

Problem setup

An object is specified to follow a straight line track with constant speed along the x-axis in a hypothetical coordinate system.
The starting position $x(0)$ is known.
The speed shall, ideally, be constant $v(t) = v(0)$, where the start speed $v(0)$ is set, i.e. known.

In discrete-time the ideal movement can therefore be expressed:

$$x(n) = x(n-1) + T_s v(n-1), \quad n = t/T_s$$
$$v(n) = v(n-1), \quad n = t/T_s$$

where T_s is the sampling time.

Unfortunately, due to a non-ideal actuator there will be a stochastic variation in the set speed during the movement.

$$v(n) = v(n-1) + w(n)$$

where $w(n)$ is white zero-mean gaussian noise

$$w(n) = N(0, \sigma_w^2)$$

During the movement, the actual speed $v(n)$ cannot be observed / measured.

The position $x(n)$ is wanted to be followed during movement and can be observed /measured, but the measurements are infected with measurement noise, so that the observed position $z(n)$ is given by

$$z(n) = x(n) + v(n)$$

where $v(n)$ is white zero-mean gaussian noise $v(n) = N(0, \sigma_v^2)$

Problem

Based on

- start conditions $x(0), v(0)$
- observed positions $z(n)$
- knowledge about the speed-noise $w(n)$
- knowledge about the measuring noise $v(n)$

it is wanted that during the movement:

- the position $x(n)$ is estimated with a (1D) Kalman tracking filter

1D tracking with 2-state Kalman filter

1) Set up a signal model and an observation model for a vector Kalman filter on the form

$$\mathrm{X(n)} = \mathrm{A(n)X(n-1)} + \mathrm{W(n)} \qquad \text{and}$$
$$\mathrm{Z(n)} = \mathrm{H(n)X(n)} + \mathrm{V(n)}$$

where the state vector is

$$\mathrm{X(n)} = \begin{bmatrix} \mathrm{x(n)} \\ \mathrm{v(n)} \end{bmatrix} = \begin{bmatrix} \mathrm{position} \\ \mathrm{speed} \end{bmatrix}$$

State the matrices $\mathrm{A(n)}, \mathrm{W(n)}, \mathrm{Z(n)}, \mathrm{H(n)}, \mathrm{V(n)}$

2) MATLAB-realisation:
   - Implement the signal model, the observation model and the Kalman vector filter algorithm
   - Use first, for instance, the following suggestion of specifications
   - $\mathrm{T_s} = 1$
   - N = 201 samples (0:200)
   - $\mathrm{X(0)} = \begin{bmatrix} \mathrm{x(0)} \\ \mathrm{v(0)} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
   - $\sigma_{\mathrm{w}}^2(\mathrm{n}) = \sigma_{\mathrm{w}}^2 = 0.01 \cdot \mathrm{v(0)}^2$
   - $\sigma_{\mathrm{v}}^2(\mathrm{n}) = \sigma_{\mathrm{v}}^2 = 0.01 \cdot (\mathrm{v(0)NT_s})^2$
   - Plot the following
   - actual, observed and estimated position
   - actual and estimated speed
   - Kalman gains
   - MSE for position and for speed
   - Comment the plots
   - Run the simulation several times due to stochastic variation.
   - Check the steady-state values by means of ***dlqe***
   -

3) Further MATLAB-experiments
   - Optionally add a control input $\mathrm{U(n)}$ to the signal model (e.g., variation in ideal speed over time)
   - Variate the measurement noise and speed noise, observe position/speed-plots and steady-state values, explain
   - Examine the result of changing the initial value $\widehat{\mathrm{X}}(0)$ for the Kalman-filter
   - Add a deterministic contribution to the speed noise, e.g. $\cos(2\pi n / N)$. Is the Kalman-filter still capable of tracking?

1D tracking with 3-state Kalman filter

Now it is wanted that the movement is to be performed along the x-axis with a constant acceleration

The state vector is extended to 3 elements $X(n) = \begin{bmatrix} x(n) \\ v(n) \\ a(n) \end{bmatrix} = \begin{bmatrix} \text{position} \\ \text{speed} \\ \text{acceleration} \end{bmatrix}$

The initial conditions $X(0) = \begin{bmatrix} x(0) \\ v(0) \\ a(0) \end{bmatrix}$ are assumed to be known / set.

During the movement the position – as previously – can be measured, however with measurement noise as previously, while the actual speed and acceleration cannot be measured.

Now assume that the stochastic variation $w(n)$ is in the set acceleration instead of as previously in the set speed.

4) Now write up, as in question 1, a signal model and an observation model for a vector Kalman filter with the 3 mentioned states.
   Specify all matrices.

5) MATLAB-realisation:
   - Implement the signal model, the observation model and the Kalman vector filter algorithm
   - Apply firstly, e.g., the following suggestions for specifications
   - $X(0) = \begin{bmatrix} x(0) \\ v(0) \\ a(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$
   - $\sigma_w^2(n) = \sigma_w^2 = 0.01 \cdot a(0)^2$
   - $\sigma_v^2(n) = \sigma_v^2 = 0.01 \cdot [v(0)NT_s + \frac{1}{2}a(0)(NT_s)^2]^2$
   - Plot the following
   - actual, observed and estimated position
   - actual and estimated speed
   - actual and estimated acceleration
   - Kalman gains
   - MSE for position, speed and acceleration
   - Comment on the plots
   - Run the simulation several times in consideration of stochastic variation.
   - Check the steady-state values by means of ***dlqe***
   - You might experiment with measurement noise, acceleration noise, control input and initial conditions.